CE444: Geotechnical Design

Homework Assignment #5

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Problem 1.

A shallow square footing is being considered as a foundation option to support both gravity and seismic loads for a marquee sign. The footing width is 4-by-4 meters and the embedment depth D = 1.5 meters. The soil deposit includes a saturated clay with undrained shear strength Su = 50 kPa and total unit weight of γ = 17 kN/m3. The water table is very deep. The mass of the marquee sign is approximately m = 100 metric ton (100 Mg). Assume pinned connection between column and foundation (i.e. no bending moment transferred to the foundation).

Find:

Part A

Determine q_{bL}^{net}.

Part B

Determine factor of safety, FS.

Solution:

Part A

The equation to determine q_{bL}^{net} is as follows.

$$q_{bL}^{net} = S_u N_c s_c i_c d_c + q_0 N_q - q_0$$

To get the factors to determine q_{bL}^{net}, the following equations were used.

$$N_c = 2 + \pi$$

$$s_c = 1 + 0.2\frac{B}{L} = 1.2 \qquad \textit{Square Footing; B = L}$$

$$i_c = 1 - 1.3\frac{Q_h}{Q_v} = 1 - 1.3\frac{438.5\ \text{kN}}{981\ \text{kN}} = 0.419 \qquad \textit{Ignore } \frac{Q_h}{Q_v} \leq 0.4 \textit{ condition.}$$

$$d_c = 1 + 0.2\frac{D}{B} = 1 + 0.2\frac{1.5\ \text{m}}{4\ \text{m}} = 1.075$$

$$N_q = 1 \qquad \textit{Clay}$$

$$q_0 = D_f\gamma = 1.5\text{m} \times 17\text{kNm}^{-3} = 25.5\ \text{kPa}$$

Given the value of S_u, we can calculate q_{bL}^{net}.

$$q_{bL}^{net} = 50\ \text{kPa}(2+\pi)(1.2)(0.419)(1.075) + 25.5\ \text{kPa}(1) - 25.5\ \text{kPa}$$

$$\boxed{q_{bL}^{net} = 137.8\ \text{kPa}}$$

Part B

The factor of safety is calculated as follows.

$$\text{F.S.} = \frac{q_{bL}^{net}}{q^{net}}$$

To calculate q^{net}, the following equation is used.

$$q^{net} = \frac{Q_v}{B^2} - q_0 = \frac{981 \text{ kN}}{(4 \text{ m})^2} - 25.5 \text{ kPa} = 35.8 \text{ kPa}$$

The value of the factor of safety can now be determined.

$$\boxed{\text{F.S.} = \frac{q_{bL}^{net}}{q^{net}} = \frac{137.8 \text{ kPa}}{35.8 \text{ kPa}} = 3.8}$$

Problem 2.

GIVEN:

A rectangular footing of width B and length 2B will need to support a vertical dead plus live load of 1.7 MN (including the expected weight of the foundation) with a factor of safety against bearing failure of 2.5. The footing base will be placed 1.4 m below the ground surface in a deep deposit of sand. The water table is at a depth of 20 m. The soil's total unit weight above the water table is 18 kN/m3 and below the water table is 20 kN/m3.

FIND:

PART A

Prepare a spreadsheet that computes the minimum footing width (B) required to provide adequate safety against bearing failure

PART B

Plot B versus the effective friction angle of the sand (for ϕ' values of 30° to 45°).

SOLUTION:

Constant values of γ = 18 kNm^{-3}, D = 1.4 m, $q_0 = \gamma D$ = 25.2 kPa, and F.S. = 2 were used.

The following equations were used to calculate the factors in **Table 1.**

$$L = 2B \qquad \textit{as given by the problem statement}$$

$$N_q = \frac{1+\sin\phi'}{1-\sin\phi'}\exp\left[\pi\tan\phi'\right]$$

$$N_\gamma = 1.5(N_q - 1)\tan\phi'$$

$$\beta = tan^{-1}\left(\frac{Q_h}{Q_v}\right) = tan^{-1}(0) = 0$$

$$i_q = \left(1 - \frac{\beta}{90°}\right)^2 = (1-0)^2 = 1$$

$$i_\gamma = \left(1 - \frac{\beta}{\phi'}\right)^2 = (1-0)^2 = 1$$

$$s_q = s_\gamma = 1 + 0.1\frac{B}{L}\frac{1+\sin\phi'}{1-\sin\phi'} = 1 + \frac{0.1}{2}\frac{1+\sin\phi'}{1-\sin\phi'}$$

$$d_q = d_\gamma = 1 + 0.1\frac{D}{B}\sqrt{\frac{1+\sin\phi'}{1-\sin\phi'}} = 1 + 0.1\frac{1.4\text{m}}{B}\sqrt{\frac{1+\sin\phi'}{1-\sin\phi'}}$$

These factors are able to be used to calculate the demand, capacity, and the factor of safety relationship.

$$q^{net} = \frac{1700\text{kN}}{2B^2} - q_0 \qquad \textit{Demand}$$

$$q_{bL}^{net} = q_0 N_q s_q i_q d_q + \frac{1}{2}\gamma N_\gamma s_\gamma i_\gamma d_\gamma - q_0 \qquad \textit{Capacity}$$

$$\text{F.S.} = \frac{q_{bL}^{net}}{q^{net}} \Rightarrow 0 = q_{bL}^{net} - \text{F.S.}q^{net}$$

By setting all variables in the above equation(s) in terms of B and ϕ', the Sympy module in Python can solve the above expression for B for each given value of ϕ' from 30 degrees to 45 degrees.

PART A

$\phi'[\circ]$	B [m]	L [m]	N_q	s_q	i_q	d_q	N_γ	s_γ	i_γ	d_γ	q_{bL}^{net}[kPa]	q^{net}[kPa]	FS
30	1.511454	3.022908	18.401122	1.150000	1	1.160433	15.069814	1.150000	1	1.160433	867.184461	346.873784	2.5
31	1.422017	2.844034	20.630793	1.156202	1	1.174013	17.693056	1.156202	1	1.174013	987.871092	395.148437	2.5
32	1.335853	2.671707	23.176776	1.162729	1	1.189068	20.786382	1.162729	1	1.189068	1127.806888	451.122755	2.5
33	1.252857	2.505714	26.092012	1.169606	1	1.205808	24.442415	1.169606	1	1.205808	1290.804155	516.321662	2.5
34	1.172928	2.345855	29.439792	1.176857	1	1.224482	28.774323	1.176857	1	1.224482	1481.601535	592.640614	2.5
35	1.095971	2.191942	33.296091	1.184509	1	1.245387	33.920950	1.184509	1	1.245387	1706.134162	682.453665	2.5
36	1.021900	2.043799	37.752497	1.192592	1	1.268877	40.053378	1.192592	1	1.268877	1971.896747	788.758699	2.5
37	0.950633	1.901266	42.919912	1.201140	1	1.295379	47.383379	1.201140	1	1.295379	2288.436718	915.374687	2.5
38	0.882097	1.764193	48.933253	1.210187	1	1.325409	56.174342	1.210187	1	1.325409	2668.031587	1067.212635	2.5
39	0.816224	1.632448	55.957459	1.219775	1	1.359602	66.755509	1.219775	1	1.359602	3126.630559	1250.652224	2.5
40	0.752955	1.505911	64.195206	1.229945	1	1.398737	79.540612	1.229945	1	1.398737	3685.180332	1474.072133	2.5
41	0.692239	1.384479	73.896891	1.240748	1	1.443780	95.052451	1.240748	1	1.443780	4371.517537	1748.607015	2.5
42	0.634032	1.268065	85.373623	1.252234	1	1.495945	113.955527	1.252234	1	1.495945	5223.109785	2089.243914	2.5
43	0.578300	1.156599	99.014259	1.264464	1	1.556767	137.099663	1.264464	1	1.556767	6291.088403	2516.435361	2.5
44	0.525016	1.050032	115.307902	1.277502	1	1.628208	165.578787	1.277502	1	1.628208	7646.281582	3058.512633	2.5
45	0.474167	0.948334	134.873841	1.291421	1	1.712808	200.810761	1.291421	1	1.712808	9388.402998	3755.361199	2.5

Table 1. Prepared spreadsheet of values used to find B given a FS = 2.5 and a variable effective friction angle of sands.

PART B

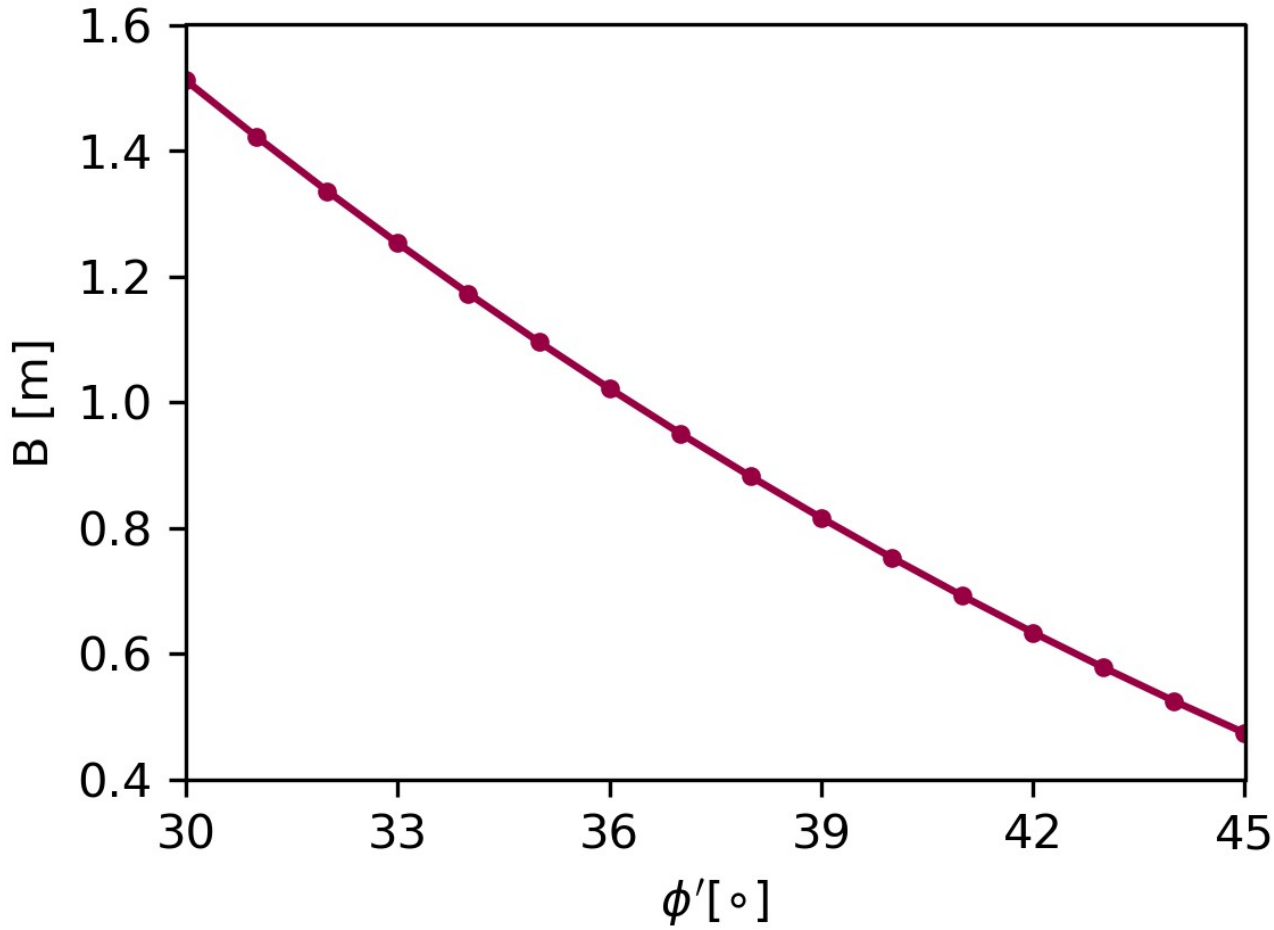

Figure 1. Required foundation width to achieve a 2.5 factor of safety with respect to effective friction angle of sand.

Problem 3.

GIVEN:

Square footings of width B will support columns for an elevated bridge that crosses over a park that periodically floods during winter. The footings will be embedded a depth Df in a deep deposit of medium-dense gravelly sand having an effective friction angle of 35 degrees. The water level can rise to about 5 feet over the ground surface. The unit weight of the gravelly sand is about 128 pcf when saturated. The design dead plus live loads imposed on each footing are a vertical load of 320 kips (along the centerline of the footing), a horizontal load of 60 kips (at the column base), and an overturning moment of 180 kip·ft (in one direction only).

FIND:

Perform all calculations in SI units.

PART A

Determine the minimum footing width B required to provide a factor of safety against bearing failure of 3.0 for embedment depths (D) ranging from 1 to 6 feet.

PART B

Plot the minimum B versus D.

SOLUTION:

PART A

Constant soil parameter values of γ' = 10.3 kNm^{-3}, $\phi' = 35°$, and F.S. = 3 were used. Constant load values of Q_v = 1423.3 kN, Q_h = 266.9 kN, and M_L = 244.0 kNm were used.

The eccentricity, e, was calculated as follows.

$$e = \frac{M_L}{Q_v} = \frac{244.0\text{kNm}}{1423.4\text{kN}} = 0.17\text{m}$$

The following equations were used to calculate the factors in **Table 2.**

$$L_{eff} = B$$

$$B_{eff} = B - 2e = B - 2(0.17\ \mathrm{m}) = B - 0.34\ \mathrm{m}$$

$$\beta = tan^{-1}\left(\frac{Q_h}{Q_v}\right) = tan^{-1}\left(\frac{266.9\mathrm{kN}}{1423.4\mathrm{kN}}\right) = 10.6^\circ$$

$$N_q = \frac{1+\sin\phi'}{1-\sin\phi'}\exp\left[\pi\tan\phi'\right] = \frac{1+\sin35^\circ}{1-\sin35^\circ}\exp\left[\pi\tan35^\circ\right] = 33.3$$

$$i_q = \left(1-\frac{\beta}{90^\circ}\right)^2 = (1-\frac{10.6^\circ}{90^\circ})^2 = 0.778$$

$$N_\gamma = 1.5(N_q-1)\tan\phi' = 1.5(33.3-1)\tan35^\circ = 33.9$$

$$i_\gamma = \left(1-\frac{\beta}{\phi'}\right)^2 = (1-\frac{10.6^\circ}{35^\circ})^2 = 0.485$$

$$q_0' = \gamma' D = 10.3\mathrm{kNm}^{-3}D$$

$$s_q = s_\gamma = 1 + 0.1\frac{B}{L}\frac{1+\sin\phi'}{1-\sin\phi'} = 1 + 0.1\ \frac{B-0.34\ \mathrm{m}}{B}\frac{1+\sin35^\circ}{1-\sin35^\circ}$$

$$d_q = d_\gamma = 1 + 0.1\frac{D}{B}\sqrt{\frac{1+\sin\phi'}{1-\sin\phi'}} = 1 + 0.1\frac{D}{B}\sqrt{\frac{1+\sin35^\circ}{1-\sin35^\circ}}$$

These factors are able to be used to calculate the demand, capacity, and the factor of safety relationship.

$$q^{net} = \frac{Q_v}{B_{eff}L_{eff}} - q_0' \qquad \textit{Demand}$$

$$q_{bL}^{net} = q_0' N_q s_q i_q d_q + \frac{1}{2}\gamma N_\gamma s_\gamma i_\gamma d_\gamma - q_0' \qquad \textit{Capacity}$$

$$\mathrm{F.S.} = \frac{q_{bL}^{net}}{q^{net}} \Rightarrow 0 = q_{bL}^{net} - \mathrm{F.S.}q^{net}$$

By setting all variables in the above equation(s) in terms of B and D, the Sympy module in Python can solve the above expression for B for each given value of D from 1 foot to 6 feet (i.e. 0.3048 m to 1.8288 m).

D [m]	q_0 [kPa]	N_q	s_q	i_q	d_q	N_γ	s_γ	i_γ	d_γ	B [m]	B_{eff} [m]	L [m]	q_{bL}^{net} [kPa]	q^{net} [kPa]
0.304800	3.139177	33.296091	1.330114	0.777931	1.020124	33.920950	1.330114	0.485225	1.020124	3.252451	2.909568	3.252451	441.811102	147.270367
0.609600	6.278354	33.296091	1.326439	0.777931	1.044546	33.920950	1.326439	0.485225	1.044546	2.971724	2.628841	2.971724	527.756773	175.918924
0.914400	9.417531	33.296091	1.322743	0.777931	1.073450	33.920950	1.322743	0.485225	1.073450	2.734353	2.391469	2.734353	624.752602	208.250867
1.219200	12.556708	33.296091	1.319030	0.777931	1.107024	33.920950	1.319030	0.485225	1.107024	2.531242	2.188359	2.531242	733.204376	244.401459
1.524000	15.695885	33.296091	1.315295	0.777931	1.145477	33.920950	1.315295	0.485225	1.145477	2.355278	2.012394	2.355278	853.821101	284.607034
1.828800	18.835062	33.296091	1.311530	0.777931	1.189067	33.920950	1.311530	0.485225	1.189067	2.200999	1.858115	2.200999	987.598397	329.199466

Table 2. Prepared spreadsheet of values used to find B given a FS = 2.5 and a variable embedded depth of foundation.

Part B

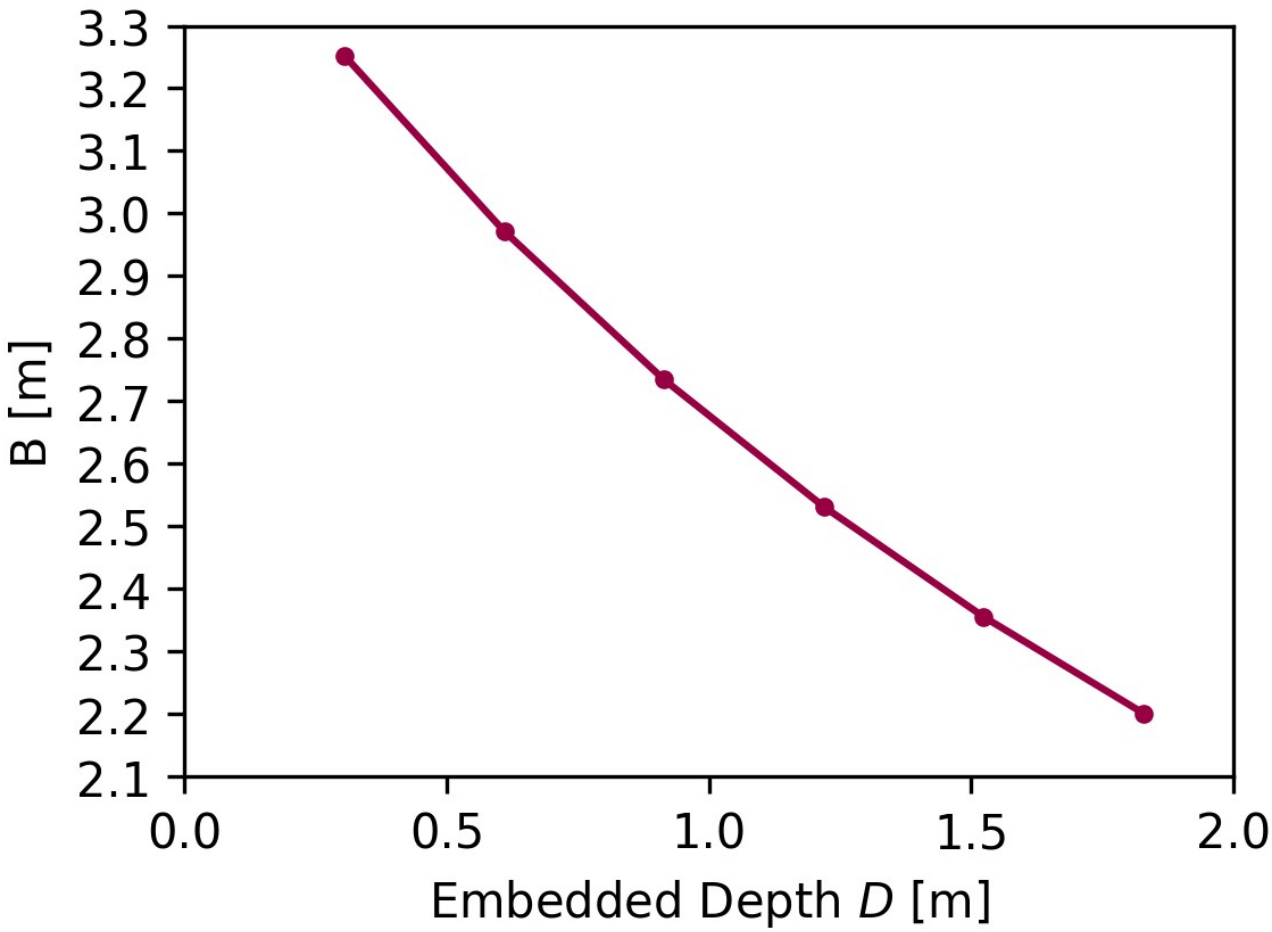

Figure 2. Required foundation width to achieve a 3 factor of safety with respect to embedded depth of foundation.